Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-5 — Properties of Logarithms

Classwork — Give it a go

Show all your working. Name the property or formula at every step. A correct answer with no working is only half an answer.

Student name :	Date : .....

Part A — Properties

Questions 1 to 4: expand and condense

1. Expand using the product rule and simplify.

$$\log_4(16x)$$

2. Expand using the quotient rule and the power rule.

$$\log_2 \frac{x^3}{8}$$

3. Condense to a single logarithm, then evaluate.

$$\log_6 4 + \log_6 9$$

4. Condense to a single logarithm.

$$2\log_5 x - \log_5 3$$

Part B — Change of base Questions 5 to 7

5. Evaluate to 3 decimal places. Write the ratio first.

$$\log_3 20$$

6. Evaluate exactly, without a calculator, by choosing a base that suits both numbers.

$$\log_9 27$$

7. Write t as a logarithm, then evaluate it to 3 decimal places.

$$4 \cdot 3^{2t} = 100$$

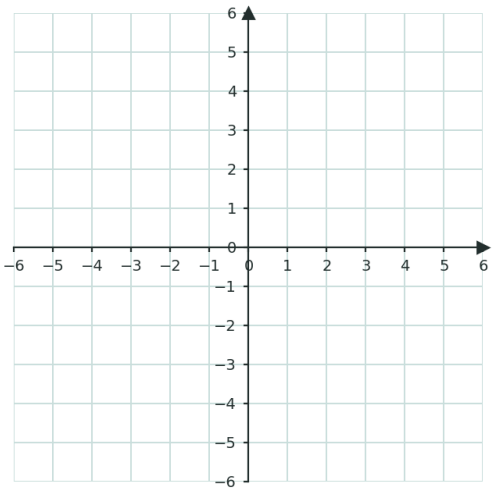
Part C — Sequences, a sketch and an error Questions 8 to 10

8. Which term is 1536? Use a logarithm, then check in the formula.

$$a_1 = 6, \quad a_{n+1} = 2a_n$$

9. Rewrite with the product rule as $2 + \log_2 x$, then plot at least three points of the graph and draw its asymptote as a dashed line.

$$y = \log_2(4x)$$



10. A student writes this. Test it with $x = 4$, then explain what is wrong.

$$\log_2(x + 4) = \log_2 x + 2$$